

Kickstarter Campaign Analysis to Predict Outcome & the Percent of Fundraising Goal to be Raised

Video Demo: <https://youtu.be/bcLjKAgVaYg>

Executive Summary

The goal of this analysis was to predict the outcome of Kickstarter campaigns prior to their start date so that Kickstarters could better optimize their marketing efforts to ensure a higher rate of success for their crowdfunded products. Additionally, as a secondary goal, we wanted to predict the percentage of the total fundraising goal that would be raised by the Kickstarter deadline. The goal of this secondary portion was to provide the Kickstarter with an additional piece of information that would help them assess by what proportion their campaign would succeed/fail and adjust their efforts accordingly to market the product, garner endorsements, explain the product in a more user-friendly manner, or any other ways they can think of.

The first stage of the analysis was to preprocess the dataset. Initial cleaning occurred via Microsoft Excel due to the fact that the comma-separated value file could be edited in its native format. The presence of empty variables complicated the import by pandas, so it was simplest to drop those variables in Excel and then move the dataset into pandas. Once in pandas, further preprocessing to drop String variables occurred. Many of the details in String variables had been encoded in their own quantitative variables elsewhere in the dataset. In order to perform correlational analysis, as I planned to, and to perform supervised learning, we needed to capture as much of the categorical information in numeric form as possible. As such, we dropped redundant String variables from the dataset. Additionally, we had to create the secondary target variable for predicting percent of fundraising goal. This was created as the ratio of the 'pledged' variable to the 'goal' variable by simple division. Missing values were dealt with by dropping the affected rows. The affected rows accounted for a fraction of 1% of the overall number of rows, so removing them would not have skewed the distributions of any of the variables. Additionally, careful consideration was put into the decision to outright drop those rows, as there was certainly an option to impute with the mean value for the column. However, upon further inspection, it was determined that imputation was going to be more costly than it was worth since the number of missing values did not exceed 20 in a dataset of over 20,000 records. It was easier to drop. Finally, categorical variables that were not already encoded in numeric variables were dummy encoded. This significantly raised the number of variables in the dataset to 73 from 66.

Moving onto the exploratory analysis, a heat-map of Spearman correlation coefficients showed some rather unsurprising associations between variables. Variables that were time-based (those encoding years, months, days, minutes, etc.)

were highly correlated with each other. In those cases, it was not surprising since the start *date* of a campaign is naturally going to be correlated to the start *year* of the campaign, as they are overlapping in the information they encode. There were many such associations. The reason these overlapping variables were not removed during this stage was because if they were not integral in explaining the variance in the data, they would eventually be removed during the dimensionality reduction stage.

Principal component analysis (PCA) was conducted for both sets of predictors and target variables (one set for each target variable). For outcome prediction, it was found that just 25 of the original 73 variables captured 95.10% of the variance in the data. And for percent of goal prediction, it was found that 23 of the original 73 accounted for 95.00% of the variance.

All of the classification and regression techniques used required normalize data, so the data were normalized using MinMax Normalization so that their values ranged from 0 to 1. For KNN Classifier, Decision Tree Classifier, SGD Regressor and KNN Regressor, simple holdout partitioning of 80% training to 20% testing samples was used. For Lasso and Ridge Regressors, 5-fold cross-validation was employed.

Both the KNN and Decision Tree classifiers performed exceptionally well on the data. KNN was performed on a range of `n_neighbors`, and the optimal model as chosen from the resulting set of models. Both KNN and Decision Tree were able to predict binary outcomes with over 99% accuracy on both training and test sets. Additionally, both models had mean absolute errors of less than 0.01. The MAE for KNN Classifier was slightly lower, however, at 0.00218 compared to Decision Tree Classifier's 0.0039. Both were exceptionally well-fit to the data, but generalized well as evidenced by the similarity between training and testing scores. Also, the high percentages of testing accuracies show they are capable of generalizing well to novel data.

As for predicting percent of fundraising goal met, the KNN Regressor fit exceptionally poorly to the data. KNN Regressor also was run on a range of `n_neighbors`. The model had a negative test accuracy on all numbers of neighbors, with the lowest accuracy being approximately -18%. According to the sklearn documentation for KNN Regressor's score function, the maximum score is 100% but there is no minimum score since the score can digress negatively, indicating an arbitrarily worsening accuracy. Clearly, this model was not the right choice for the dataset and others were tried. The mean absolute error, however was not abysmal at 1.57119, though still high.

SGD Regressor produced the lowest mean absolute error at 1.29448, but with very low training and testing accuracies of less than 0.1% each. The model was fit with a range of alphas, penalties and l1-ratios. And the final model that was chosen had an alpha of 0.5, utilized L2 regularization and, by extension had an l1-ratio of 0.10% of samples. Just for comparison, I tried ordinary least squares for multivariate regression and ended up with a 8.89 mean absolute error, so I knew that I could eliminate the simplest model and would have to proceed to more complex offerings.

Ridge and lasso regressors performed worse than SGD Regressor, with mean absolute errors of 9.45743 and 5.07089, respectively. Both models also were run on a variety of alpha values in order to choose the optimal set of parameters. And both models resulted in training and testing accuracies less than 0.01%.

The best model out of the four (or five if you could ordinary least squares) was SGD Regressor. This is obviously not saying much for the predictive power of the model or the generalizability in real-world use, since the accuracy was less than chance for each.

When I think back to why the regression models performed so poorly, I can imagine one major reason. As was discussed in the introduction of the project notebook, the overwhelming majority of Kickstarter campaigns fail with almost no pledges received. So the target values for this portion of the analysis are heavily skewed right. Any model fit to this data would be heavily biased to predict 0% pledged even when the label contradicts this. The solution to this problem is to obtain more data. However, that was not feasible as more recent datasets simply do not measure for nearly as many variables as this one. As such, obtaining newer data would be highly tedious and difficult, and would take time, making it fall outside of the scope of this assignment.

All in all, the analysis was successful for predicting outcome but unsuccessful for predicting the percent of goal pledged. Some ideas for future work may include obtaining data for Kickstarter campaigns from February 1st, 2017 (when this dataset ends) to present day (late November, 2018). Then plot the distribution of the target variable, `pct_of_goal`, to see if the distribution has shifted and become less skewed.